

Question number	Answer	Additional guidance	Mark
5(a)	An explanation that includes two of the following points: • by {mutations / change in (DNA) base sequence} (1) • because errors (in DNA) can occur during {DNA replication / mitosis} (1) • that {affect the functioning of the cell / result in uncontrolled cell division} (1)	ACCEPT carcinogens / named carcinogen / epigenetics / switching {on / off} genes / binding of (different) transcription factors to a gene / (mutation in) {tumour suppressor gene / proto-oncogene} ACCEPT rapid cell division / apoptosis inhibited	(2)

Question number	Answer	Mark
5(b)(i)	The only correct answer is C <i>A is incorrect because Ebola has RNA</i> <i>B is incorrect because Ebola has RNA</i> <i>D is incorrect because Ebola has RNA</i>	(1)

Question number	Answer	Additional guidance	Mark
5(b)(iv)	An explanation that includes the following points: - because {it / virus} will {burst / lyse / ruptures} the cancer cells (1) - when the newly-formed virus particles are formed / during the lytic cycle / when virus replicates (1) OR (virus infected cancer cell) will stimulate the immune system (1) - so that T killer cells can destroy the cancer cell (1)	DO NOT ACCEPT lysogenic cycle / latency / replication occurring whilst viral DNA is incorporated into cancer cell DNA ACCEPT CD8 cells / cytotoxic cells	(2)

Question number	Answer	Additional guidance	Mark
5(c)	An explanation that includes four of the following points: • {(virally-) infected / cancer} cells act as antigen-presenting cells (to T killer cells) (1) • T killer cells release {perforins / chemicals / enzymes} (1) • that will cause the {cancer / virus infected} cells to {burst / lyse} (1) • T helper cells produce cytokines to activate B cells (1) • because antibodies (to the cancer cells / virus) will be produced (by plasma cells) (1) • that will increase phagocytosis (of tumour cells by macrophages) (1)	ACCEPT T killer cells will be activated by {cancer antigens / viral antigens (on cancer cells)} stimulated / clonal selection / proliferation / produced {CD8 / cytotoxic} cells for T killer cells ACCEPT {CD8 / cytotoxic} cells IGNORE destroy ACCEPT CD4 cells for T helper cells ACCEPT resulting in opsonisation / agglutination IGNORE easier	(4)